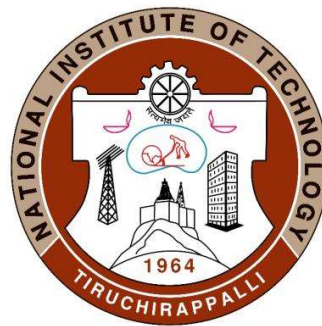


Lecture 1

Introduction to Polymers



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Properties of Engineering materials

1. Physical properties

Density

Melting Point

2. Thermal Properties

Thermal conductivities

Thermal expansion

Heat-distortion temperature

3. Mechanical Properties

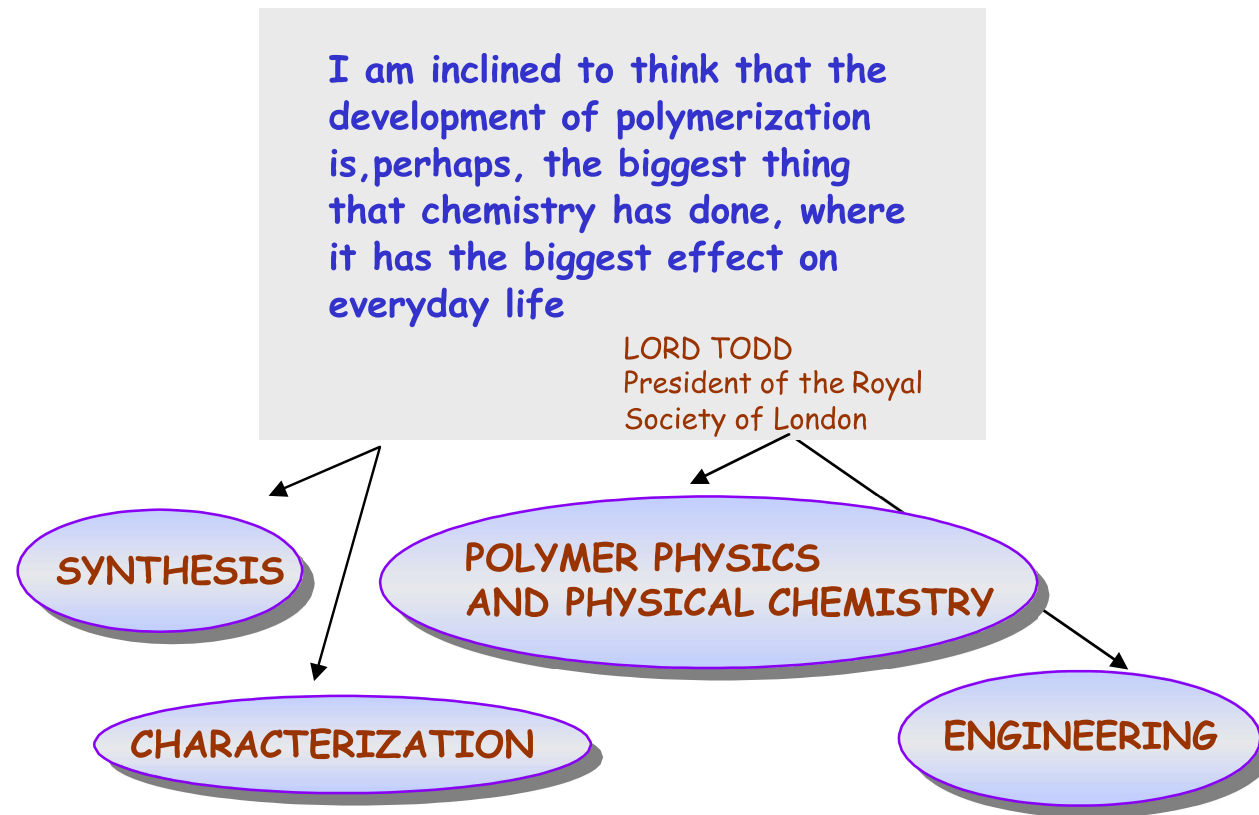
4. Electrical properties

5. Magnetic properties

6. Optical properties

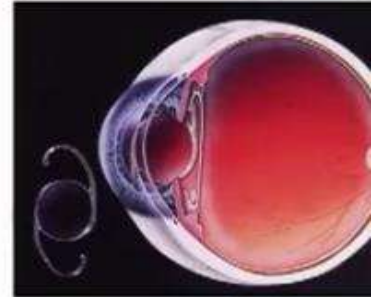
POLYMER SCIENCE AND ENGINEERING - WHAT IS IT ?

- THE SCIENCE OF LARGE MOLECULES





A collage of various medical supplies and equipment. On the left, three syringes are arranged vertically. In the center, there is a clear plastic drip chamber connected to a coiled white tube. To the right of the tube is a nebulizer mask and a small nebulizer unit. On the far right, a blue rectangular container, possibly a nebulizer solution bottle, is visible. The background is dark and textured.



POLYMERS

- THE TERM “POLYMERS” MEANS “MANY (*POLY*) PARTS (*MERS*)”.
- Polymers are defined as the high molecular weight organic materials made up of long chain molecules containing principally hydrogen and carbon and may contain atoms of N, Cl, F, O or S.
- These are complex and giant molecules with molecular weight in the range 10^4 - 10^5 and formed by many chemically bonded parts or units.

Polymers: Materials consisting of polymer molecules that consist of repeated chemical units ('mers') joined together, like beads on a string. Some polymer molecules contain hundreds or thousands of monomers and are often called macromolecules.

DUE TO HIGH MOLECULAR WEIGHT, THEY ARE ALSO KNOWN AS HIGH POLYMERS.

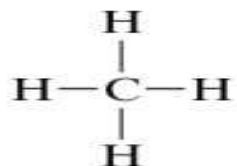
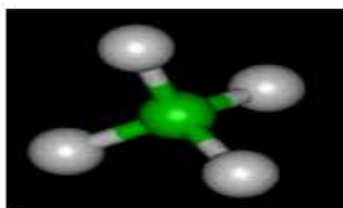
Polymers may be natural, such as leather, rubber, cellulose or DNA, or synthetic, such as nylon or polyethylene.

Silk fiber is produced by silkworms in a cocoon, to protect the silkworm while it metamorphoses in a moth.

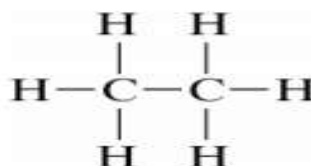
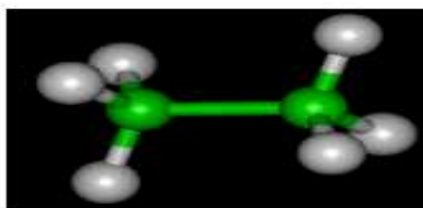
Hydrocarbon Molecule (I)

- Most polymers are organic in their origin and are formed from hydrocarbon molecules
- Each C atom has four e^- that participate in bonds, each H atom has one bonding e^-

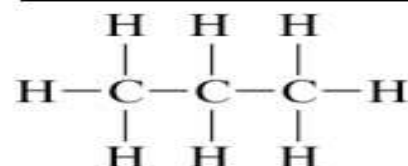
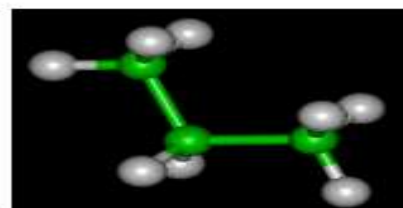
Examples of saturated (all bonds are single ones) hydrocarbon molecules



Methane, CH_4



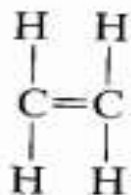
Ethane, C_2H_6



Propane, C_3H_8

Hydrocarbon Molecules (II)

Double and triple bonds can exist between C atoms (sharing of two or three electron pairs). Molecules with double and triple bonds are called unsaturated. Unsaturated molecules are more reactive

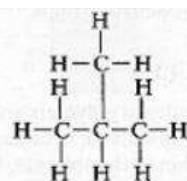
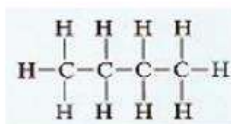


Ethylene, C_2H_4



Acetylene, C_2H_2

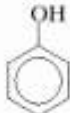
Isomers are molecules that have the same composition (contain the same atoms) but have different atomic arrangement. An example is butane and Iso butane. Physical properties (e.g. boiling temp) depend on the isomeric state.



Butane $\rightarrow \text{C}_4\text{H}_{10} \leftarrow$ Isobutane

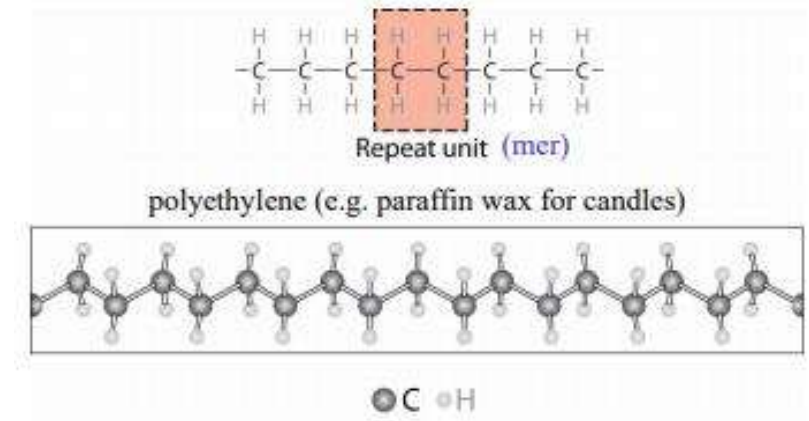
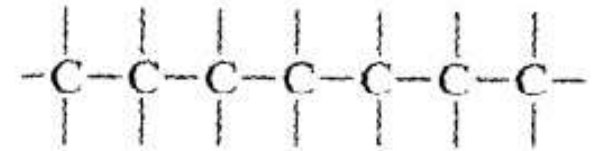
Hydrocarbon Molecule (III)

Many other organic groups can be involved in polymer molecules. In table below R represent a radical, an organic group of atoms that remains as a unit and maintain their identity during chemical reactions (e.g. CH_3 , C_2H_5 , C_6H_5).

<i>Family</i>	<i>Characteristic Unit</i>	<i>Representative Compound</i>
Alcohols	$\text{R}-\text{OH}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H} \end{array}$ Methyl alcohol
Ethers	$\text{R}-\text{O}-\text{R}'$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ Dimethyl ether
Acids	$\begin{array}{c} \text{OH} \\ \\ \text{R}-\text{C} \\ \\ \text{O} \end{array}$	$\begin{array}{c} \text{H} \quad \text{OH} \\ \quad \\ \text{H}-\text{C}-\text{C} \\ \quad \\ \text{H} \quad \text{O} \end{array}$ Acetic acid
Aldehydes	$\begin{array}{c} \text{R} \\ \\ \text{C}=\text{O} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{C}=\text{O} \\ \\ \text{H} \end{array}$ Formaldehyde
Aromatic hydrocarbons		 Phenol

Polymer Molecules

- Polymer molecules can be very large (macromolecules)
- Most polymers consist of long and flexible chains with a string of C atoms as a backbone
- Side-bonding of C atoms to H atoms or radicals
- Double bonds are possible in both chain and side bonds
- Repeat unit in a polymer chain (unit cell) is a mer.
- Small molecules from which polymer is synthesized is monomer. A single mer is sometimes also called a monomer.



Polymers are classified as:

- (1) Man-made or synthetic polymers that are synthesized in the laboratory; (nylon, poly-ethylene, poly-styrene)
- (2) Biological polymer that are found in nature. Biological polymers: DNA, proteins, carbohydrates

Homopolymer: A polymer prepared from a single monomer

A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-AA--A-

B-B-B-B-B-B-B-B-BB-B-B-B-B-B-B-B-B-B-B

Copolymer: If two or more different monomers are employed.

—A—B—A—B—A—B—A—B—A—B—A—B—A—B—

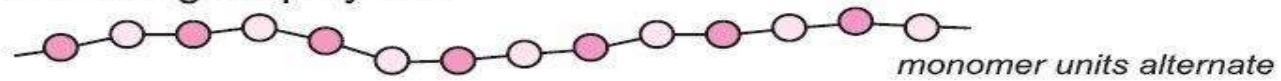
alternating copolymer

—A—A—B—A—B—B—A—B—A—A—B—B—B—A—

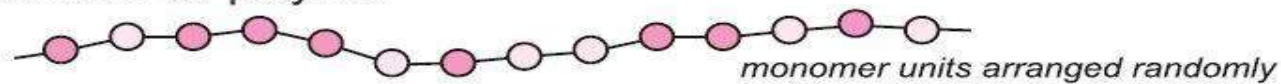
random copolymer

Homopolymer & Copolymer

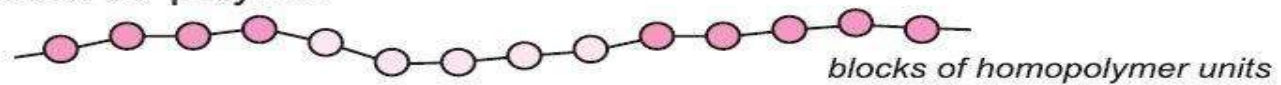
Alternating co-polymer



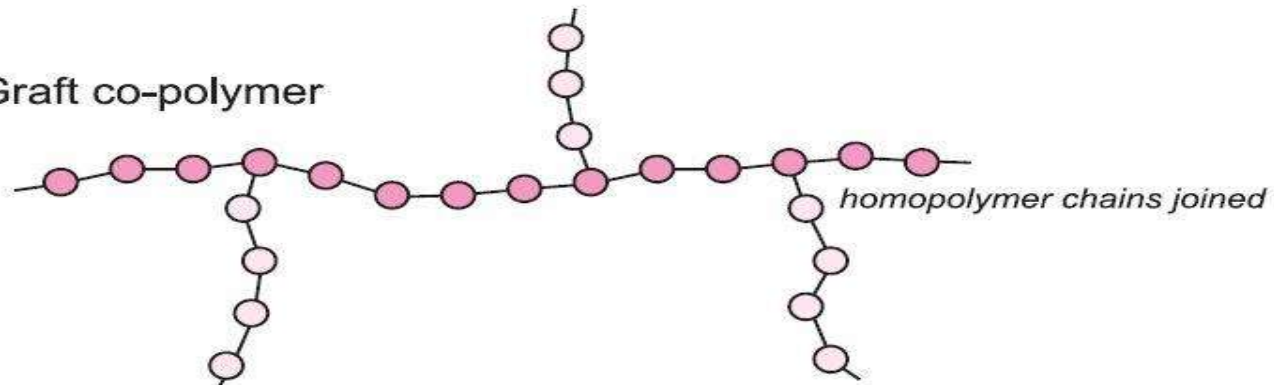
Random co-polymer



Block co-polymer



Graft co-polymer



Molecular structure

Covalent chain configuration and strength

